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Chapter 11
One Side of the Street: Half-Street Games

We begin our investigations into poker toy games by considering a class of simplified games, which we call ***half-street games.*** Half-street games have the following characteristics:

- The first player (conventionally called X) checks in the dark.
- The second player (conventionally called Y) then has the option to either check or may bet some amount according to the rules of the game.
- If Y bets, X always has the option to call, in which case there is a showdown, and may additionally have the option to fold (but not raise). If Y checks, there is a showdown.

We will make reference to the ***value*** of the game. This is the expectation for the second player, Y, assuming both sides play optimally. We will also only consider money that changes hands as a result of the betting in our game. This value is the ***ex-showdown*** value. This can include bets and calls that X and Y make on the streets of our game. It can also include the swing of a pot that moves from one player to the other as the result of a successful bluff.

The reason that we use ex-showdown value in studying these games is that we're not particularly concerned with the value to the player of action that occurred before our game. We're trying to capture the value of the betting that takes place on the half-street we are examining. The fact that the pot is five or ten or eighteen bets isn't important except as it relates to the way that the action plays in our game. We should also note that Y can't have a negative value in a half-street game, since he can simply check behind and achieve a value of zero.

Half-street toy games have at least one feature that is exceptionally valuable; we can solve them - in fact, we can solve just about any half-street game - as the complexity remains manageable throughout. As we will see, multi-street games often turn out to be intractable, or we can only obtain approximate answers. Being able to solve a game is of great benefit, as we can use the solution to gain insight into more difficult problems.

The first game we consider is a quite simple game involving a concept called ***clairvoyance*** - throughout Part III, players will be referred to as ***partially clairvoyant*** if they can see their opponents' cards (but not cards to come, if there are any), and simply ***clairvoyant*** if they can see both then opponent's cards and the cards to come. In this example, there are no cards to come, so either definition serves.

Example 11.1 - The Clairvoyance Game

One half-street.
Pot size of P bets.
Limit betting.
Y is clairvoyant.

Y's concealed hand is drawn randomly from a distribution that is one-half hands that b cat X's hand and one-half hands that do not.

In this game, X and Y each have only one decision to make. This is a common element of all limit half-street games; Y must decide on a range of hands to bet, while X must decide on a range of hands with which to call. X's "range" in this game is going to consist of all the same hand, since he is only dealt one hand. This doesn't pose a problem, however - betting and calling distributions can just as easily contain mixed strategies with particular hands as they can pure

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strategies.
We can immediately see that Y has the informational advantage in this game; he can value bet all his winning hands (call these hands ***the nuts)*** and bluff some of his losing hands (call these ***dead hands)*** with perfect accuracy.

The ex-showdown payoff matrix for this game is as follows:

	Player X	**Check-Call**	**Check-Fold**
Player Y			
Nuts	Bet	+1	0
	Check	0	0
Bluff	Bet	-1	+P
	Check	0	0

In finding the optimal strategies for this game, we should keep in mind some of the principles from the last chapter. First, we should consider pure strategies. Presume that Y value bets all his strong hands but checks all his weak ones. Then X can exploit him by folding all the time. If X does this, however, Y can switch to betting all his hands (value bets and bluffs). If Y does that, X switches to calling all the time, in which case Y can exploit him by value betting all his strong hands and checking weak ones.

So we have the familiar pattern of oscillating pure strategies; this indicates that the optimal strategies will be mixed. There are two strategy choices - one for X (how often to call a bet from Y), and one for Y (how often to bluff). Y should always bet his nut hands because this option dominates checking. X will call with some fraction of his hands. We will call this fraction *c.* Likewise. Y will bluff with some fraction of his dead hands. We will call this fraction *b.* If we find values for *b* and *c,* then we will have the optimal strategies for the game.

We can consider each of these decisions in turn.

First we consider how often X must call. When X plays his optimal strategy, Y will be indifferent to bluffing (that is, checking and bluffing his dead hands will have the same expectation). When bluffing, Y wins *p* bets by bluffing successfully (if X folds) and loses one bet when X calls. Recalling that *c* is the calling frequency, and 1 - *c* is the frequency that X folds, we have:

(pot size) (frequency X folds) = (bluff bet)(frequency X calls)
$P(1 - c) = c$
$c = P/(P+1)$

Notice that as the pot size grows, the calling ratio grows. This is in some sense an extension of the pot odds principle; the more there is in the pot, the more often X must call to keep Y from bluffing.

Y, likewise, must bluff often enough to make X indifferent to calling or folding. When calling, X loses 1 bet by calling a value bet, and gains *P* + 1 bets by calling a bluff. If *b* is the ratio of bluffs to bets, then we have:

$1 = (P+ 1)b$
$b = 1/(P+ 1)$

This value 1/(*P*+ 1) is an extremely important value in analyzing poker. It is so important that we

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assign it its own Greek letter, α (alpha).

$$\alpha = 1/(P+1) \quad \text{(limit cases)} \qquad \textbf{(11.1)}$$

α represents two things in this game. First, X must call enough to make Y indifferent to bluffing and checking his weak hands. X's calling frequency is $P/(P+1)$, or 1 - α. Therefore α is X's ***folding frequency*** when faced with a bet from Y. In addition, Y bluffs with α of his dead hands. Since he will bet with 100% of his nut hands, this makes α equal to the ***ratio of bluffs to value bets*** to keep the opponent indifferent between betting and calling.

Likewise, the ratio 1 - α is of some interest:

$$1 - \alpha = 1 - 1/(P+1) \qquad \textbf{(11.2)}$$
$$1 - \alpha = P/(P+1) \qquad \text{(limit cases)}$$

Even though we have only introduced cases with limit betting, we can generalize the expression of α to include games with different bet sizes:

For games with variable bet sizes:

$$\alpha = s/(1+s) \quad \text{(variable bet sizes)} \qquad \textbf{(11.3)}$$

where s is the bet size in pots. (For example, betting $50 into a $100 pot makes $s = {}^1/_2$)

Notice that here, as the pot size grows, α, the bluffing ratio becomes smaller: that is. Y bluffs less often into larger pots. This may seem counter-intuitive to you as there is more to gain by successfully bluffing into a large pot, but this is an important principle of optimal play:

Bluffing in optimal poker play is often not a profitable play in and of itself. Instead, the combination of bluffing and value betting is designed to ensure the optimal strategy gains value no matter how the opponent responds. Opponents who fold too often surrender value to bluffs while opponents who call too often surrender value to value bets.

We can now state that the optimal strategies for this game are:

Solution:

Y bets all his nut hands and bluffs α of his dead hands, or α/2 of his total hands.
X calls with 1 - α of his hands total.

The ex-showdown expectation of the game to Y is as follows. He wins 1 bet when he value bets and gets called. Recall that he holds half nut hands and half dead hands; hence, his value betting frequency will be ${}^1/_2$. He wins the entire pot when he bluffs successfully, and loses 1 bet when he bluffs and gets called. We know that X will call frequently enough to make him indifferent to bluffing. His EV from bluffing is therefore zero and his expectation from the game is:

$\langle Y \rangle$ = (frequency of value betting) (one bet) (frequency of getting called)
$\langle Y \rangle = ({}^1/_2)(P/(P+1))$
$\langle Y \rangle = P/2(P+1)$

Notice that as the pot grows. X's disadvantage grows; he must pay off the second player's value bets more and more often, and the number of bluffs that Y is required to make in order to keep X

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calling is smaller and smaller. Also note that Y cannot have expectation of a half bet, no matter how large the pot is - this is because Y would have to be called all the time when he value bet to have such a value.

The important features of the clairvoyance game are the existence of betting to bluffing and calling ratios, their dependence on the pot size, and the idea that Y's ability to bet is worth more as the pot grows. Changing the size of the pot and seeing how it affects strategy is a useful tool in understanding poker. We can even examine what happens when we make the pot extremely large, such that neither player will fold any hand with any value at all.

The [0,1] Distribution

We introduce here for the first time the ***[0,1] distribution.*** Throughout Part III, we discuss various games using this distribution. In [0,1] games, instead of discrete cards, both players are "dealt" a uniformly random real number between 0 and 1. This means that each player has an equal chance of being dealt any number between 0 and 1. These numbers function for these players as poker hands. The convention that we use is that if there is a showdown, then the lowest hand wins (this simplifies calculations). That means that 0 and numbers approaching it are the strongest hands, while 1 and numbers approaching it are the weakest.

Strategies for this type of game have a different structure than the strategy for the clairvoyant game. The primary way in which they are different is that in most cases, mixing is not strictly required. Since there are an infinite number of different hands, mixing on any particular hand is irrelevant, and for any interval, we can play more effectively with pure strategies, separated by a ***threshold.*** These are points along the [0,1] continuum that mark the boundaries between one action and another,

When solving [0,1] games, we will often employ the following technique:

1) Guess at the structure of the solution,
2) Solve the game as if the guessed structure were the correct one,
3) Verify that our guess was correct by showing that neither player can improve his or her strategy unilaterally.

We use the term ***parameterization*** to refer to a guess at the structure of the solution of a game. For example, one possible solution to a hypothetical [0,1] game might involve a player Y betting his best hands between 0 and y_1, checking his intermediate hands between y_1 and y_0, and betting his worst hands between y_0 and 1. A different parameterization might be that Y bets his intermediate strength hands while checking his best and worst hands. Many structures of a solution are possible; if we guess at a parameterization, we can then solve the game for that parameterization only.

Once we have identified a parameterization, we will find the optimal solution for that particular parameterization. We generate equations based on principles of being indifferent at all the thresholds. Instead of the indifference for hands that utilize mixed strategies (as we saw before), we find that at the thresholds between one action and the next, the player will be indifferent.

It may not be obvious why this is true; here is an intuitive argument. First, we know that as we move very small distances along the line (from one hand to a very slightly stronger hand) that our showdown equities are continuous - that is, the showdown equity of one hand is very close to the showdown equity of its close neighbors. If this is true, then the value of playing hands must be continuous also, since a hand's value will be composed of its equity when there's a showdown plus its equity when there is no showdown (which is constant). For example, the value of check-

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calling with the hand 0.6 should be very close to the value of check-calling with the hand 0.60001. Now consider the threshold value. Suppose the equity of one option at the threshold is higher than the equity of the other option. Then in the region of the lower-valued option, we can find a very small region very close to the threshold where we could switch from the lower-valued threshold to the higher-valued one and gain equity. If we can do this, the strategy isn't optimal. Hence optimal strategies are indifferent at the thresholds.

In most of our parameterizations, there are a fixed number of thresholds (between different strategies). For each threshold, we write an equation that relates the strategy elements that force that threshold to be indifferent. We call these ***indifference equations***. By solving these systems of equations, we can solve for each of the threshold values and find the optimal strategy for a given parameterization.

In some cases, a better strategy can be found that utilizes a different parameterization. Often, if we try to solve the equations generated from a particular parameterization that is incorrect, we encounter inconsistencies (impossible thresholds, for example). Therefore, it is necessary for us to confirm the strategies after solving for the solution to a particular parameterization in order to show that it is indeed the best one.

Example 11.2 - [0,1] Game #1

This very simple game consists of a single half-street of betting with no folding allowed, where player X is forced to check and must call Y's bet, if Y chooses to bet. For this type of game (where no folding is allowed) the pot size is irrelevant.

X has no decisions to make. Y's strategy consists of a single decision - to bet or not to bet? Y knows X's response will be to call any bets, so Y can simply bet all hands that have positive or zero expectation and check all hands with negative expectation.

We will frequently create tables such as the follow to show various types of outcomes as we have done at other points in the text.

Y's EV of betting a hand of value *y* is:

X's band	**Result**
[0, y]	1 (Y bets and X calls with a better hand)
[y, 1]	+1 (Y bets and X calls with a worse hand)

Recall that X's chance of having any hand is uniformly distributed, so the probability of each line of the table is equal to the size of the interval on which X holds that hand.

<Y, bet > = *p*(X has a better hand)(-l) + *p(*X has a worse hand)(+l)
<Y, bet > = (y – 0)(-l) + (l – y)(l)
<Y, bet > = 1 – 2y

Now we find all the hands where Y's EV is greater than 0:

1 - 2y > = 0
Y < = $^1/_2$

So Y should bet the best half of his hands; X is forced to call. When X has a hand in the range [0, $^1/_2$] Y breaks even overall, and when X has a hand in [$^1/_2$,1], Y gains 1 bet.

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Half the time, Y bets, and half that time, he gains a bet, making his overall equity from this game $^1/_4$.

Viewing this graphically:

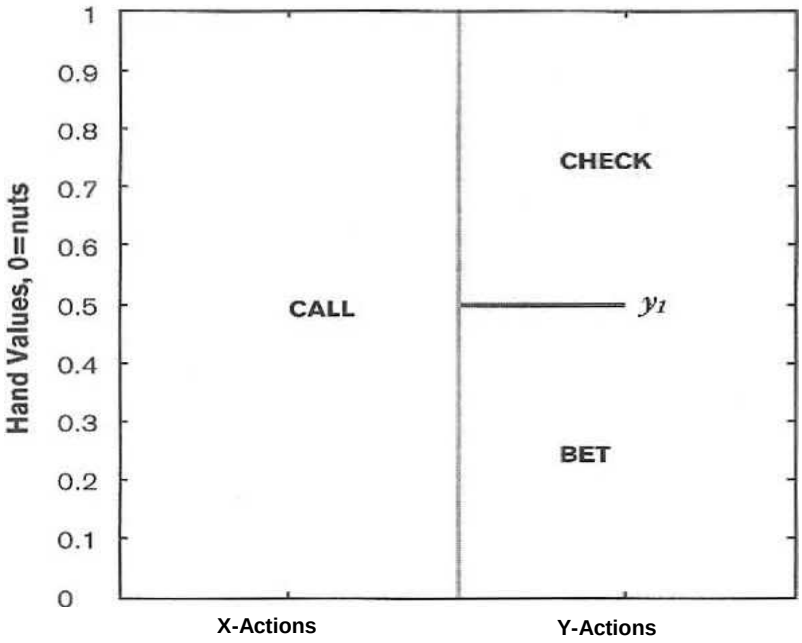

Figure 11.1 **Game** #1--**Half Street, no fold**

The no-folding game is a very simple one, but it introduces the idea of [0,1] games, as well as illustrating some key principles. The first of these principles is that if the opponent has no decisions to make, simply maximizing expectation against his static strategy is optimal. The second is that if you hold a distribution similar to your opponent's, you cannot be raised, and your opponent must always call, you should bet half your hands. We will see this in all types of cases where there is no more betting.

Example 11.3 - [0,1] Game #2

In the preceding example, we made a special rule that does not normally apply to poker situations - that no folding was allowed. We found that in this "no fold" game there was no bluffing - the betting region for Y was simply his top half of hands by value. Poker is often perceived to be fundamentally a game of bluffing, and so we turn our attention to games where one or the other player may fold.

[0.1] Game #2 is identical to Game #1 (Example 11.2), except that instead of being forced to call, X may instead fold. It is in this case that the pot size again becomes relevant, as X's decision about whether to call depends on the value he stands to gain by calling. We use the convention (as we generally will in limit poker) of calling the pot size P and the size of the bet 1 unit.

In Game #1, we expressed Y's strategy as a single threshold; however, as games become more complex, we will have multiple thresholds to consider. We will assign these values to variables in a consistent way:

- x_n as the threshold between strategies which put in the nth bet and strategies which only put in the (n - l)th bet for value. (When more bets and check-raising are allowed, we will see that the threshold x_2, for example, is a threshold between checking and betting

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initially.)

- x_0 as a specific threshold (the threshold between bluffing and checking).
- X_n* as the threshold between calling the nth bet and folding to it.

By the first of these rules, the Game #1 strategy for y would be expressed as y_1, since Y is betting hands above that for value. There is no y_0 value, or perhaps it can be thought of as being at 1. Since X was forced to call all the time, x_1* was also 1.

In Game #2, X actually has a strategy, and it is made up of a single threshold value x_1*, which separates the region where X calls from the region where X folds to a bet from Y. The calling region for X will be an interval [0, x_1*], and the folding region an interval [x_1*, 1].

We can show that Y plays a strategy that includes some bluffs as follows:

If X and Y are playing optimally, x_1* will be chosen such that it maximizes expectation against Y's strategy. Let's see what happens if Y uses a strategy similar to his strategy from Game #1 - using a betting region from 0 to some threshold y_1 and a checking region from y_1 to 1.

X's best response is to call with hands that have positive equity against Y's betting range. X's calling threshold x_1* will then be chosen so that X's equity is positive.

X's equity is determined by his calling threshold. If he calls at x, he will beat Y's hand and win the pot when Y is between y_1 and x_1*, and lose a bet when Y is between 0 and x_1*. So his overall equity is:

<X> = (pot value when he wins)(chance of winning) - (one bet) (chance of losing)
$\langle X \rangle = (P+1)(y_1 - x_1^*) - 1(x_1^*)$

When this is positive, he will call.

$(P+1)(y_1 - x_1^*) - 1(x_1^*) > 0$
$(P+1)(y_1) - (P+2)\, x_1^* > 0$
$x_1^* > y_1\,(P+1)/(P+2)$

So X would call with a fraction of Y's betting hands, such that the ratio of Y's bets to X's calls would be $(P+1)/(P+2)$,

Intuitively, this makes sense. Let's say that the pot is 1, so the bet is pot-sized. Then X would need to have a $^1/_3$ chance of having the best hand in order to call. The threshold such that he would have such a chance is $^2/_3$ of the way from 0 to y_1.

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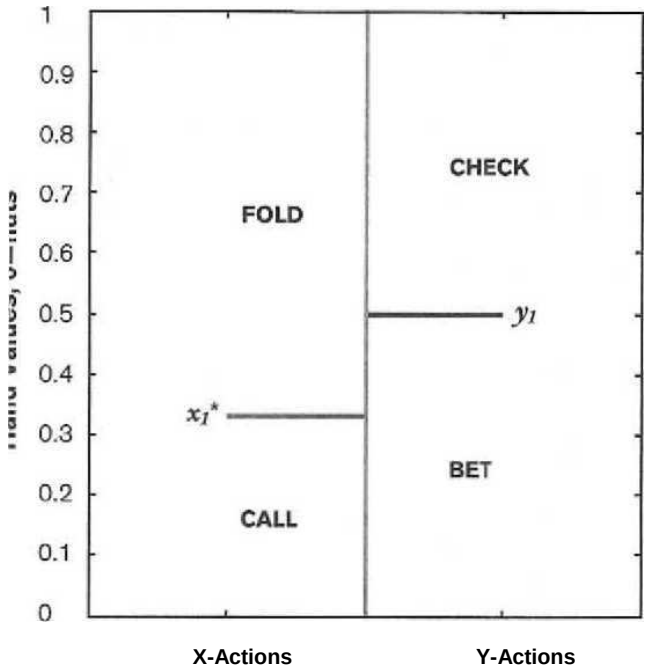

Figure 11.2. Anon-optimal strategy for Y in [0,1] Game#2

However, if X played this way, Y could unilaterally improve his equity by checking hands from x_1 to y_1 and instead bluffing an equal amount of hands near 1. By doing this, he improves the equity of those bluffs because X will more often fold the winning hand against those bluffs than against the hands on x_1, y_1.

Since Y can unilaterally improve his equity by changing his strategy, then his strategy cannot be optimal. But the assumption that led to this was that Y's strategy was optimal. So something must be wrong with our assumption that Y should only value bet. Y must have a bluffing region.

We can parameterize Y's strategy as follows:

Y bets in a region of strong hands from 0 to some threshold y_1.
Y checks in an intermediate region from y_1 to some threshold y_0.
Y bets weak hands in a region from y_0 to 1.
We also guess for this parameterization that x_1* lies to the right of y_1.

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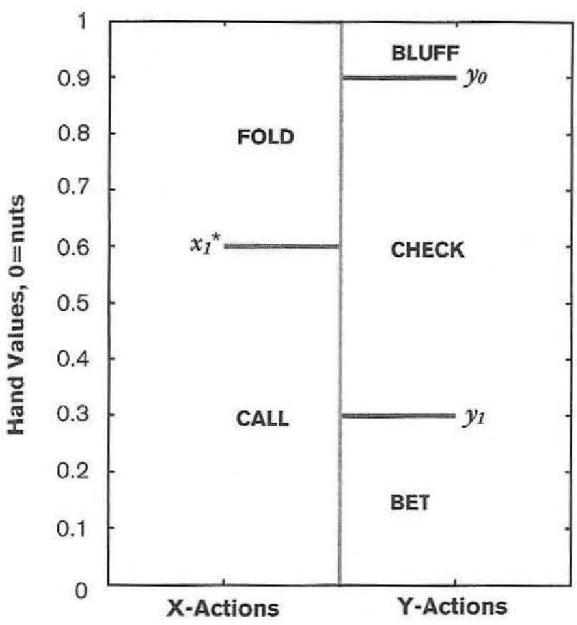

Figure 11.3. Optimal strategy for [0,1] Game #2

We know that if X and Y are playing optimal strategies, then at x_1*, X will be indifferent between calling and folding.

The indifference at x_1* (X indifferent between calling and folding):

Y's hand	*p*(Y's hand)	<X, call>	Product	<X, fold>	Product
$[0, y_1]$	y_1	-1	$- y_1$	0	0
$[y_0,1]$	$1 - y_0$	$P + 1$	$(P + 1)(1 – y_0)$	0	0
Total			$\mathbf{- y_1 + (P + 1)(1 – y_0)}$		0

Note that we omit Y's hands on $[y_1, y_0]$ as with those hands he has not bet and so we are not faced with this decision. The sums of the "Product" columns must then be equal - these are the weighted average equities across all of Y's hands.

$- y_1 + (P + 1)(1 – y_0) = 0$
$y_1 = (P + 1)(1 – y_0)$
$1 – y_0 = y_1 (1/(P + 1))$

$$1 – y_0 = \alpha y_1 \qquad \textbf{(11.4)}$$

In our parameterization, $1 – y_0$ represents the length of the interval of bluffing hands, just as y_1 represents the number of hands that will be value bet. The relationship between these two quantities is the same as it was in the clairvoyant game - the betting to bluffing ratio is still α.

We can then look at the indifference at the Y's two thresholds.

The indifference at y_1 (Y indifferent between betting and checking):

X's hand	*p*(X's hand)	<Y, bet>	Product	<Y, check>	Product
$[0, y_1]$	y_1	-1	$- y_1$	0	0
$[y_1, x_1*]$	$x_1* - y_1$	+ 1	$x_1* - y_1$	0	0
$[x_1*, 1]$	$1 - x_1*$	0	0	0	0
Total			$x_1* - 2y_1$		0

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Setting the expected values equal:

$x_1^* - 2y_1 = 0$

$y_1 = x_1^* / 2$ **(11.5)**

The indifference at y_0 (Y indifferent between betting and checking):

X's hand	*p*(X's hand)	<Y, bet>	Product	<Y, check>	Product
$[0, x_1^*]$	x_1^*	-1	$- x_1$	0	0
$[x_1^*, y_0]$	$y_0 - x_1^*$	$+ P$	$(y_0 - x_1^*)P$	0	0
$[y_0, 1]$	$1 - x_1^*$	0	0	0	0
Total			$\boldsymbol{Py_0 - (P + 1)\, x_1^*}$		0

$Py_0 - (P + 1)\, x_1^* = 0$

Recalling from Equation 11.3 that $1 - y_0 = \alpha y_1$, we have:

$P(1 - \alpha y_1) = (P + 1)\, x_1^*$
$x_1^* = P/(P+1)\,(1 - \alpha y_1)$

Remembering that $(1 - \alpha) = P/(P + 1)$, we have:

$x_1^* = (1 - \alpha)\,(1 - \alpha y_1)$ **(11.6)**

This result is of equal importance as it determines the calling ratio in continuous cases. To make Y indifferent to bluffing at y_0, X must call with a fraction of his hands that can beat a bluff equal to P/(P + 1), or (1 - α). Alternatively, we can say that X can fold α of his hands that can beat a bluff and that will make Y indifferent to bluffing.

Combining the three indifference equations, we can find a solution to the game.

$x_1^* = (1 - \alpha)\,(1 - \alpha y_1)$ **(11.6)**

$y_1 = x_1^* / 2$ **(11.5)**

$x_1^* = 2y_1$
$2y_1 = (1 - \alpha y_1)\,(1 - \alpha)$
$2y_1 = 1 - \alpha y_1 - \alpha + \alpha 2y_1$
$1 - \alpha = 2y_1 + \alpha y_1 - \alpha 2y_1$
$1 - \alpha = y_1(2 - \alpha)(\alpha + 1)$
$y_1 = (1 - \alpha) / (2 - \alpha)(\alpha + 1)$

Since $x_1^* = 2y_1$:

$x_1^* = 2\,(1 - \alpha) / (2 - \alpha)(\alpha + 1)$
$1 - y_0 = \alpha y_1$
$1 - y_0 = \alpha\,(1 - \alpha) / (2 - \alpha)(\alpha + 1)$

This game was one of the games solved by John von Neumann and Oskar Morganstern in their seminal work *Gam Theory and Economic Behavior* (1944).

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We see in this game a number of important points. First, we see that X's ability to fold or call changes the basic structure of Y's strategy. In Game #1, Y simply bet his best hands for value. However, in Game #2, if Y employs such a strategy, X simply folds all his weaker hands, and Y never makes any money. Y's response is to begin to bluff his weakest hands, forcing X to call with hands between Y's value bets and Y's bluffs in order to keep Y from bluffing.

In this game, we also see α play a prominent role. Just as in the clairvoyant game, X folds α of his hands that can beat a bluff, making Y indifferent to bluffing, and Y makes his ratio of bluffs to bets α to make X indifferent to calling with hands between y_1 and y_0.

A third important idea here is the following. What is the value of bluffing? In the clairvoyant game, the answer was zero; if Y checked, he simply lost the pot. In this game, however, the answer is not quite so simple. At y_0, Y is indifferent between bluffing and checking - that is, the situation is the same as the clairvoyant game (from an ex-showdown perspective). But at, for example, 1, Y's equity from bluffing is the same as his equity at y_0 (since X will only call with hands above x_1*). So Y is not indifferent to bluffing at 1 - in fact, for all the hands he bluffs which are worse than y_0, he increases his equity by doing so!

In this chapter, we presented two very simple half-street games and a third that was more complex. In the clairvoyant game, we saw that the clairvoyant player bluffs and value bets hi a precise ratio optimally, while the player whose hand is exposed calls and folds hi a related ratio to make the clairvoyant player indifferent to bluffing. In [0.1] Game #1, we introduced the [0,1] distribution and saw that when one player has no strategic options, the other player simply maximizes his equity. In [0,1] Game #2, we saw a similar pattern of betting best hands and bluffing worst hands, and we saw the effect that bluffs have on the calling threshold, forcing X to call with hands worse than Y's worst value bet.

Key Concepts

- Value betting and bluffing in optimal strategies are directly related to each other in half-street games by the ratio $\alpha = 1(P+1)$. By bluffing in this ratio, the bettor makes the caller indifferent to calling.
- α is also the proper folding ratio for X in half-street games. By folding this ratio, the caller makes the bettor indifferent to bluffing.
- When one player has no strategic options, the other player should simply maximize his expectation.
- Balancing value bets with bluffs properly ensures that a player extracts value from the opponent no matter how the opponent plays; if he folds too often, the bettor gains on his bluffs, and if he calls to often, the bettor gains on his value bets.
- Optimal strategies are not always indifferent to bluffing; only the threshold hands are truly indifferent. Often the weakest bluffs have positive value (compared to checking).

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